

SYDNEY BOYS' HIGH SCHOOL

MOORE PARK, SURRY HILLS

TRIAL HIGHER SCHOOL CERTIFICATE EXAMINATION

1996

MATHEMATICS

2/3 UNIT

*Time allowed - Three hours
(Plus 5 minutes reading time)*

Examiners: R.Boros, D.Hespe

DIRECTIONS TO CANDIDATES

- * ALL questions may be attempted.
- * The marks allocated to each question are indicated.
- * All necessary working should be shown in every question. Full marks may not be awarded for careless or badly arranged work.
- * Standard integrals are printed at the back. Approved calculators may be used.
- * Each section attempted is to be returned in a *separate* bundle, clearly marked Section A (Q1, Q2), Section B (Q3, Q4), Section C (Q5, Q6), Section D (Q7, Q8), or Section E (Q9, Q10). Each bundle must also show your name. Start each question on a new page.
- * If required, additional paper may be obtained from the Examination Supervisor upon request.

NOTE: This is a trial paper only and does not necessarily reflect the content or format of the Higher School Certificate Examination Paper for this subject.

SECTION A (Hand up separately)

Question 1 (start a new page)

(2 marks) [a] Factorise and simplify $\frac{2x^2 - 8}{x^3 - 8}$

(1 mark) [b] Solve $\frac{x-1}{2} + \frac{x+2}{3} = 5$

(3 marks) [c] Solve $|2x-1| > 5$ and graph your solution on a number line.

(2 marks) [d] If $f(p) = \begin{cases} p^2 & \text{for } p < 2 \\ 3p-2 & \text{for } 2 \leq p < 3 \\ \frac{1}{3}p^3-2 & \text{for } p \geq 3 \end{cases}$
evaluate $f(-1) + f(2) + f(5)$

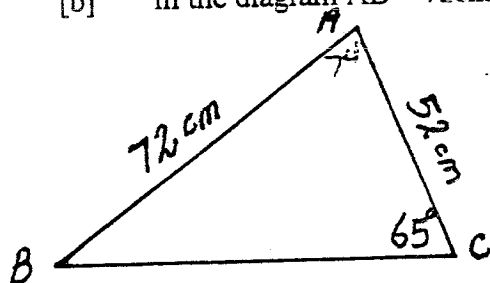
(2 marks) [e] Write down the exact value of $\cot 60^\circ$

(2 marks) [f] Express $\frac{1+\sqrt{3}}{1-\sqrt{3}}$ in the form $a + b\sqrt{3}$
where a and b are rational numbers.

Question 2 [start a new page]

(2 marks) [a] Find the perpendicular distance from the origin to the line whose equation is $3x - 4y = 30$. Hence give the equation of the circle, centre the origin, to which the line $3x - 4y = 30$ is a tangent.

[b] In the diagram $AB = 72\text{cm}$, $AC = 52\text{cm}$ and $\hat{ACB} = 65^\circ$



[figure not to scale]

(2 marks) (i) Find the size of $\hat{ABC}$ to the nearest degree.

(2 marks) (ii) Hence find the area of triangle ABC to 2 significant figures.

Question 2(Cont.)

[c] Given the points A(-4, 2), B(0, -4), C(1, 1)

(2 marks)

(i) Find the equation of the line AB, written in the general form.

(2 marks)

(ii) A line passing through the point C, perpendicular to AB, meets AB at N. Show that N has coordinates $(-\frac{2}{5}, -\frac{1}{5})$

(2 marks) [d] Solve for θ if $0^\circ \leq \theta \leq 360^\circ$, $3\tan^2\theta = 1$

SECTION B (Hand up separately)

Question 3 [start a new page]

[a] Differentiate with respect to x:

(1 mark)

(i) $2x^2 - 4x + 9$

(1 mark)

(ii) $(4 - 5x)^3$

(2 marks)

(iii) $\frac{e^{2x}}{\sin x}$

(2 marks)

(iv) $2x/n(3-x)$

[b] Find

(1 mark)

(i) $\int (\sin 2x - 2\sec^2 x) dx$

(1 mark)

(ii) $\int_1^2 (3x + 1)^2 dx$

(2 marks)

(iii) $\int_1^3 (\frac{1}{x+1} - \frac{1}{x^2}) dx$ (leave your answer in exact form)

(2 marks)

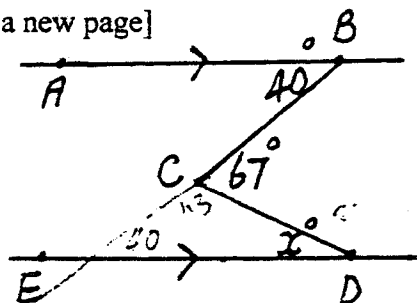
[c] The gradient function of a curve is given by $\frac{dy}{dx} = 3 - 4x$.

Find the equation of the curve if it passes through the point (6, -5)

Question 4 [start a new page]

(1 mark)

[a]

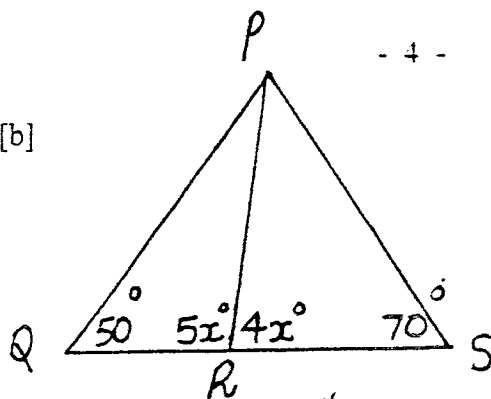


In the diagram $AB \parallel ED$.
Find x , giving reasons.

3 marks)

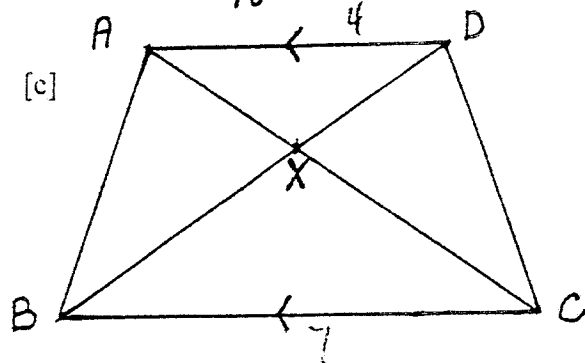
[b]

Given QRS is a straight line,
prove that PR bisects the angle QPS.



[c]

ABCD is a trapezium, $AD \parallel BC$.
Intervals AC and BD intersect at X.



(2 marks)

(i) Prove that $\frac{AX}{AD} = \frac{CX}{BC}$,

(1 mark)

(ii) Given $AD = 4\text{cm}$, $BC = 7\text{cm}$ find the ratio AX to CX.

(2 marks)

[d] Prove that $\frac{1 + \cos \theta}{\sin \theta} + \frac{\sin \theta}{1 + \cos \theta} = 2 \operatorname{cosec} \theta$.

3 marks)

[e] Shade on a number plane the region for which
 $y \leq \sqrt{16 - x^2}$ and $y \geq 2$ hold.

SECTION C (hand up separately)

Question 5

[start a new page]

(2 marks)

[a] An urn contains 7 green discs and 4 blue discs. Two discs are withdrawn at random, one at a time, the first one not being replaced before the second one is drawn. Find the probability that the two discs drawn are of different colours.

[b]

A particle is moving in a straight line so that its distance t seconds after the initial observation is x cm from a fixed point O. The position of the particle is given by $x = t^3 - 6t^2 + 9t$.

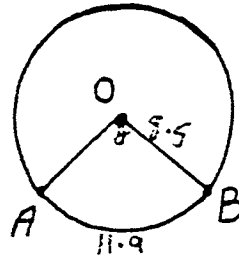
Question 5 [b] Cont.

- (2 marks) (i) Find the initial position and initial velocity of the particle.
- (2 marks) (ii) Show that the particle is momentarily at rest on two occasions.

[c] An arc of length 11.9cm subtends an angle of θ at the centre O of a circle of radius length 8.5cm. Find:

- (2 marks) (i) the value θ in both radians (to 1 decimal place) and degrees (nearest degree),

- (2 marks) (ii) the area of sector AOB.



- (2 marks) [d] Draw a neat sketch of the curve $y = 3\sin 2x$ for the domain $0 \leq x \leq 2\pi$.

Question 6 [start a new page]

[a] At the beginning of each year a man invests \$2400 in a superannuation scheme which offers a 10.5% p.a. return on his investment. He contributes to this scheme for 24 years. The interest on his investment is compounded every year. Find to the nearest dollar:

- (1 mark) (i) the value of his first \$2400 investment at the end of the 24 year period.
- (3 marks) (ii) the total value of the superannuation at the end of the 24 year period.

[b] Given the parabola $y = x^2 - 2x - 3$.

- (2 marks) (i) express this equation in the form $(x-h)^2 = 4a(y-k)$.
- (1 mark) (ii) state the coordinates of the vertex.
- (1 mark) (iii) state the coordinates of the focus.
- (1 mark) (iv) what is the equation of the directrix?

Question 6 [b] Cont.

- (1 mark) (v) what is the equation of the normal to this parabola at the point where this parabola crosses the y axis?

[c] If α and β are the roots of $2x^2 - 4x + 9$ what is the value of

- (1 mark) (i) $\frac{1}{\alpha} + \frac{1}{\beta}$
- (1 mark) (ii) $(\alpha - \beta)^2$

SECTION D (hand up separately)

Question 7 (start a new page)

- (2 marks) [a] Use the trapezoidal rule with 3 function values to find an approximate value for

$$\int_0^{\frac{\pi}{2}} \sin^2 x \, dx$$

Answer correct to 3 significant figures.

- (4 marks) [b] Determine the area enclosed by the curve $f(x) = 3x(2 - x)$ and the x axis from $x = 1$ to $x = 3$.

- (4 marks) [c] Find the volume generated when the region bounded by the curve $y = x^2 + 1$ and the x axis between $x = 0$ and $x = 5$ is rotated around the x axis. Leave your answer in exact form in terms of π .

- (2 marks) [d] Find the value of x if $\log x^2 - \log 2x = \log 8$. All logs are to the same base.

Question 8 (start a new page)

- (2 marks) [a] Sketch the graph of a function given that

$$f(2) = 0, f'(2) = 0$$

$$f'(x) < 0 \text{ for all } x < 2$$

$$f'(x) > 0 \text{ for all } x > 2.$$

- (3 marks) [b] A plant is observed over a period of time. Its initial height is 40cm. It grows 6cm during the first week of observation. Each succeeding week the growth is 70% of the previous weeks growth. Assuming this pattern continues, calculate the plant's ultimate height.

- [c] Consider the curve $y = x^3 + x^2 - x + 1$.

- (3 marks) (i) Find any turning points and determine their nature.
- (2 marks) (ii) Find any points of inflexion
- (2 marks) (iii) Sketch the curve for $-2 \leq x \leq 1$

SECTION E (hand up separately)

Question 9 (start a new page)

- (2 marks) [a] Solve $2^x \times 4^{x+1} = 0.5$

- (3 marks) [b] Find the value(s) of k for which $x^2 - (k-2)x + (k+1) = 0$ has real roots.

- (3 marks) [c] Find the locus of a point $P(x,y)$ which moves so that it is equidistant from the point $S(-4, 0)$ and the y axis.

Question 9 [d]Cont.

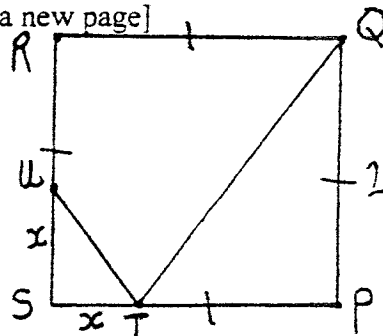
- [d] A particle starts from rest at 0 and moves in a straight line with a velocity V cm/s after t seconds, given by $V = 16t - 4t^2$.

(2 marks) (i) Find the acceleration after 2 seconds.

(2 marks) (ii) Find the distance travelled in the first 2 seconds.

Question 10 [start a new page]

[a]



PQRS is a square of side length 2cm.
ST = SU = x cm as shown on the diagram.

- (1 mark) (i) Show that the area A cm², of the quadrilateral QRUT is given by

$$A = 2 + x - \frac{1}{2}x^2$$

(2 marks) (ii) Find the value of x for which A is a maximum.

(1 mark) (iii) Hence determine the maximum area for QRUT.

- [b] Ron borrows \$95000 to buy a cottage. He is charged interest on the balance owing at the rate of 10.5% p.a. compounded monthly and agrees to repay the loan including interest making equal monthly instalments of \$M.

(1 mark) (i) How much does Ron owe at the end of the first month just before he makes an instalment?

(1 mark) (ii) Write an expression involving M for the total amount owed by Ron just after the first instalment is paid.

Question 10 [b] Cont.

- (3 marks) (iii) Calculate the value of M (to the nearest cent) which will repay the loan after 25 years.
- (3 marks) (iv) In how many months (to the nearest whole month) will the loan be repaid if Ron made instalments of \$1500 per month?

THIS IS THE END OF THE PAPER

STANDARD INTEGRALS

$$\int x^n dx = \frac{1}{n+1} x^{n+1}, \quad n \neq -1; \quad x \neq 0, \text{ if } n < 0$$

$$\int \frac{1}{x} dx = \ln x, \quad x > 0$$

$$\int e^{ax} dx = \frac{1}{a} e^{ax}, \quad a \neq 0$$

$$\int \cos ax dx = \frac{1}{a} \sin ax, \quad a \neq 0$$

$$\int \sin ax dx = -\frac{1}{a} \cos ax, \quad a \neq 0$$

$$\int \sec^2 ax dx = \frac{1}{a} \tan ax, \quad a \neq 0$$

$$\int \sec ax \tan ax dx = \frac{1}{a} \sec ax, \quad a \neq 0$$

$$\int \frac{1}{a^2 + x^2} dx = \frac{1}{a} \tan^{-1} \frac{x}{a}, \quad a \neq 0$$

$$\int \frac{1}{\sqrt{a^2 - x^2}} dx = \sin^{-1} \frac{x}{a}, \quad a > 0, \quad -a < x < a$$

$$\int \frac{1}{\sqrt{x^2 - a^2}} dx = \ln \left(x + \sqrt{x^2 - a^2} \right), \quad x > a > 0$$

$$\int \frac{1}{\sqrt{x^2 + a^2}} dx = \ln \left(x + \sqrt{x^2 + a^2} \right)$$

NOTE: $\ln x = \log_e x, \quad x > 0$

(1)(a) $\frac{2(x+2)}{x^2+2x+4}$

(b) $x = 4\frac{3}{5}$

(c) $\{x > 3 \text{ or } x < -2\}$

(d) $34\frac{2}{3}$

(e) $\frac{\sqrt{3}}{3}$

(f) $-\sqrt{3} - 2$

(2)(a)(i) $3\sqrt{10}$ (ii) $x^2 + y^2 = 90$

(b)(i) 41^0 (ii) 1800 cm^2

(c)(i) $3x + 2y + 8 = 0$ (ii) Proof

(d) $30^0, 150^0, 210^0, 330^0$

(3)(a)(i) $4(x-1)$ (ii) $-15(4-5x)^2$

(iii) $\frac{e^{2x}(2\sin x - \cos x)}{\sin^2 x}$

(iv) $2\ln(3-x) - \frac{2x}{3-x}$

(b)(i) $-\frac{1}{2}\cos 2x - 2\tan x + c$ (ii) 31

(iii) $\ln 2 - \frac{2}{3}$

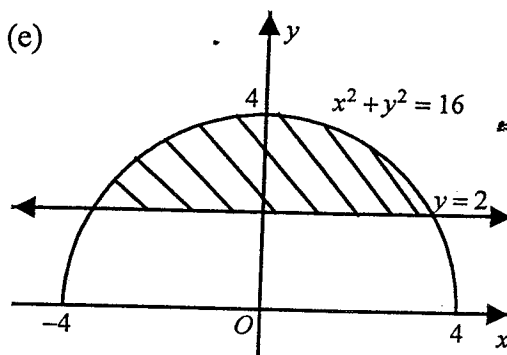
(c) $y = -2x^2 + 3x + 49$

(4)(a) $x = 27$ (b) Proof

(c)(i) Proof (ii) $\frac{4}{7}$

(d) Proof

(e)



(5)(a) $2 \times \frac{4}{11} \times \frac{7}{11} = \frac{56}{121}$

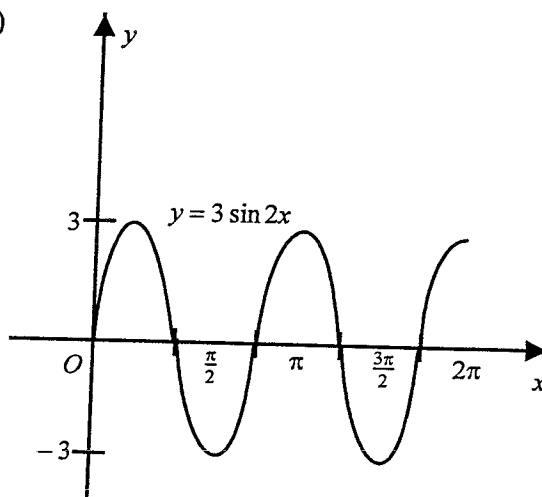
(b)(i) $x = 0, v = 9 \text{ cm/s}$

(ii) Let $v = 0$

(c)(i) $1.4 \text{ rad} \approx 80^0$

(ii) 50.575 units^2 (to 3 d.p)

(d)



(6)(a)(i) \$26 358 (to the nearest dollar).

(ii) \$281 250

(b)(i) $(x-1)^2 = y+4$

(ii) $(1, -4)$

(iii) $\left(1, -3\frac{3}{4}\right)$

(v) $x - 8y - 24 = 0$

(c)(i) $\frac{4}{9}$ (ii) -14

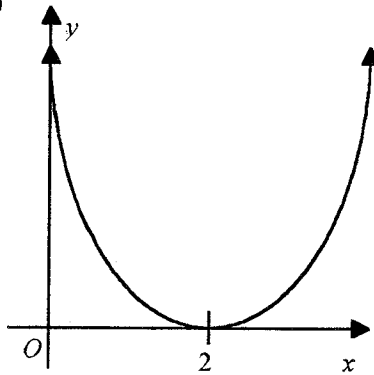
(7)(a) 0.785 (to 3 s.f.)

(b) 6 units^2

(c) $755\pi \text{ units}^3$

(d) $x = 16$

(8)(a)

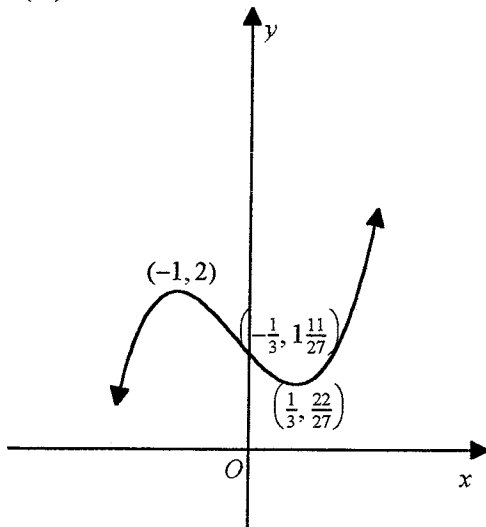


(b) 60

(c)(i) $(-1, 2)$ rel. max.; $(\frac{1}{3}, \frac{22}{27})$ rel. min.

(ii) $(-\frac{1}{3}, 1\frac{11}{27})$

(iii)



(9)(a) $x = -1$

(b) $x \geq 4 + 2\sqrt{2}$ or $x \leq 4 - 2\sqrt{2}$

(c) Locus is a parabola : $y^2 = 8x + 16$

(d)(i) $a = 0 \text{ cm/s}^2$ (ii) $21\frac{1}{3} \text{ cm}$

(10)(a)(i) Proof

(ii) $x = 1$

(iii) Area = 2.5 cm^2

(b)(i) $\$95831.25$

(ii) $\$(95831.25 - M)$

(iii) $\$896.97$

(iv) $93 \text{ months (to the nearest month)}$